

Day 168/365

Keeping the curve honest

Polynomial regression on messy data, explained with everyday examples.

Think of a student before an exam

Student A understands the topic. Scores 80 on last year's paper and 78 on a new one.

Student B memorizes last year's answers. Scores 100 on that paper and 40 on a new one.

A model with too many bends is Student B. It looks perfect on the data it studied, then fails on new data.

Today's goal: build a model that behaves like Student A.

Example: price of a used car by age

Say we have 10 cars. We try three curves:

Curve	Score on our 10 cars	Guess for a 12-year-old car
Too straight (degree 1)	Okay	A bit off
Gentle bend (degree 2)	Good	Sensible
Very wiggly (degree 9)	Perfect	Absurd, even a negative price

The wiggly curve touches all 10 points, so it looks like the best one. That's the trap. The gentle bend is the one you can trust.

Illustrative example, not real data.

Fix 1: Charge a penalty for wiggles

This is Ridge, the same idea as my car price series. Imagine a teacher who deducts marks for needlessly complicated answers.

alpha is how strict the teacher is:

alpha	What the curve does
0 (no teacher)	Wiggles freely to touch every point
Small to medium	Bends only where the data really insists
Huge	Too scared to bend, almost a flat line

So the right alpha sits in the middle, like a good pair of brakes: not off, not locked.

Fix 2: Put features on the same scale

A car aged 10 years gives us these columns:

age	age ²	age ³
10	100	1,000
20	400	8,000

It's like measuring one thing in kilometres and another in millimetres. The penalty would treat them unfairly.

Fix: standardize the columns, so they're all in comparable units, before applying Ridge.

I used this same idea in the Day 161 app's standardize toggle.

Fix 3: Don't peek at the test data

Imagine setting the exam rules after seeing everyone's answer sheets. That's peeking.

Same thing in code:

Wrong: standardize the whole dataset, then split it. The test data already shaped the scaling.

Right: split first. Learn the scaling from the training part only, then apply it to the test part.

Put the steps (curve, scaling, Ridge) in one pipeline and this happens automatically inside every cross-validation fold.

It's the same family of mistake as my Day 160 leakage bug.

Keeping the curve honest

Let unseen data choose the settings

Try a few degrees and alphas. For each one, record the error on the folds it did **not** train on:

Degree	alpha	Error on unseen data
1	1	0.40
3	1	0.22 ✓
9	0.01	0.65
9	10	0.25

Pick the lowest error. Notice that degree 9 with a strong alpha also does well, so Ridge can rescue a high degree. If two settings tie, choose the simpler one.

Illustrative numbers.

3 rules to remember

1. Penalize wiggles.
2. Scale before you penalize.
3. Never peek at the test data.

Be Student A, not Student B.

Which one has bitten you most: overfitting, scaling or leakage?